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Paper Code : OEC-IT701A/OECCS701A/OECD5 701A Operations Research

UPID : 007549

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (I) In waiting line theory, number of customers waiting in the queue is referred to as
a) Queuing system b) Traffic intensity c) Queue length d) System length
- (II) For finding an optimum solution in transportation problem _____ method is used.
a) Simplex b) Big-M c) MODI d) Hungarian
- (III) In PERT, the maximum time that is required to perform the activity under extremely bad conditions is known as
a) Normal time b) Optimistic time c) Most likely time d) Pessimistic time
- (IV) A game is said to be fair if the value of the game is
a) One b) Two c) Three d) Zero
- (V) In North-west corner method, first allocation is made at
a) Lower right corner of the table
b) Upper right corner of the table
c) Highest cost cell of the table
d) Upper left-most corner of the table
- (VI) The assignment problem always form a _____ matrix.
a) circular b) square c) rectangular d) triangular
- (VII) The difference between the maximum time available and the actual time needed to perform an activity is known as
a) Free float b) Independent float c) Total float d) Half float
- (VIII) When all the players of the game follow their optimal strategies, then the expected pay-off of the game is called
a) Gain of the game b) Loss of the game c) Value of the game d) Pay of the game
- (IX) Traffic intensity in Queuing theory is also called
a) Service factor b) Arrival factor c) Utilization factor d) Consumption factor
- (X) An assignment problem is considered as a particular case of
a) Transportation problem b) Sequencing problem c) Queuing problem d) Game theory
- (XI) 'Buffer Stock' is _____
- (XII) The point of intersection of a pure strategy game is called
a) Value of game b) Two person game c) Mixed strategy game d) None of these

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. Solve the following LPP using graphical method: [5]
Maximize $Z = 3x_1 + 2x_2$;
Subject to $x_1 \leq 4$; $2x_2 \leq 12$; $3x_1 + 2x_2 \leq 18$; $x_1 \geq 0$; $x_2 \geq 0$.
3. A small maintenance project consists of the following jobs whose precedence relationships with duration (days) are given below: [5]

Jobs	1-2	1-3	2-3	2-5	3-4	3-6	4-5	4-6	5-6	6-7
Days	15	15	3	5	8	12	1	14	3	14

Find the critical path and total project duration.

4. Find the saddle point and optimal strategies having the following pay-off matrix: [5]

	B1	B2	B3
A1	10	2	3
A2	7	6	8
A3	0	3	1

5. Draw a network diagram for the following project. Find critical path and project duration. [5]

Activity	A	B	C	D	E	F	G	H	I	J	K	L	M
Predecessor	-	A	B	A	D	E	-	G	J, H	-	A	C, K	I, L
Duration (days)	6	4	7	2	4	10	2	10	6	13	9	3	5

6. Solve the LPP by graphical method [5]

$$\text{Maximize } Z = 3x_1 + 2x_2$$

$$\text{Subject to } 2x_1 + x_2 \leq 2$$

$$3x_1 + 4x_2 \geq 12$$

$$x_1, x_2 \geq 0$$

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. Find the critical path for the following problem. The following table shows the jobs of a network along with their time estimates (in days). [15]

Job	1-2	1-6	2-9	2-4	3-5	4-5	5-8	6-7	7-8
to	3	2	6	2	5	3	1	3	4
tm	6	5	12	5	11	6	4	9	9
tp	15	14	30	8	17	15	7	27	28

- i) Draw the project network. <https://www.makaut.com>
 ii) Find the critical path.
 iii) Find the probability that the project is completed in 31 days.
8. Solve the following assignment problem using Hungarian method. [15]

	M1	M2	M3	M4
J ₁	1	4	6	3
J ₂	9	7	10	9
J ₃	4	5	11	7
J ₄	8	7	8	5

9. Solve the following LPP by Simplex method [15]

$$\text{Maximize } Z = 60x_1 + 50x_2$$

$$x_1 + 2x_2 \leq 40$$

$$3x_1 + 2x_2 \leq 60 ; \quad x_1, x_2 \geq 0$$

10. Solve the following LPP by Simplex method [15]

$$\text{Minimize } Z = 4x_1 + 8x_2 + 3x_3$$

$$x_1 + x_2 \geq 2$$

$$2x_1 + x_3 \geq 5 ; \quad x_1, x_2, x_3 \geq 0$$

11. Solve the following transportation problem using VAM method, check optimality using MODI method and find the optimal solution. [15]

Destination Source	D ₁	D ₂	D ₃	Supply
S ₁	10	7	8	45
S ₂	15	12	9	15
S ₃	7	8	12	40
Demand	25	55	20	

*** END OF PAPER ***